The Effect of Early Warning System Probation on Undergraduate Student Success: A Regression Discontinuity and Instrumental Variables Approach

I. Research Questions and Hypotheses

Question 1: What is the effect of being placed on academic probation by an Early Warning System (EWS) on student academic performance and retention?
Question 2: What is the effect of completing the mandatory advisor intervention on these outcomes?
Question 3: Do these effects differ by race/ethnicity and gender?

H1: The evidence on whether early warning interventions improve undergraduate student outcomes is mixed. EWS-triggered interventions have been linked to improvements in course grades, retention rates, and graduation rates, including among underrepresented students (Jayaprakash et al., 2014; Renick, 2020). At the same time, academic probation, a common intervention triggered by such systems, has been found to discourage some students from re-enrolling, and short-term GPA gains tend to fade over subsequent semesters (Fletcher & Tokmouline, 2017; Lindo et al., 2010). We hypothesize that EWS-triggered probation will have a modest positive effect on next-semester GPA and credit completion among students who remain enrolled, and we remain open to the possibility that it may also discourage some students from re-enrolling.
H2: Research on academic advising shows that students who meet regularly with an advisor are more likely to stay enrolled and make progress toward their degrees (Bettinger & Baker, 2014; Swecker et al., 2013). We hypothesize that completing the mandatory advisor meeting will have a positive effect on next-semester GPA, credit completion, and one-year retention. If the advisor meeting is the component that drives improvement, we also expect this effect to be larger than the Q1 estimate, which will be diluted by students who were flagged but never attended the meeting.
H3: Prior research has found that the effects of academic probation vary by gender (Lindo et al., 2010). At the same time, researchers have raised concerns that algorithmic prediction systems in education may produce unequal outcomes across racial groups (Baker & Hawn, 2022; Ekowo & Palmer, 2017; Kizilcec & Lee, 2022). Because of this, we hypothesize that the effects of EWS-triggered probation will differ by race/ethnicity and gender, but we are uncertain in what direction.

II. Primary Source of Data

We will use the Mid-Atlantic Public University Student Success Database (MAPU-SSD). The MAPU-SSD is a longitudinal administrative dataset that tracks undergraduate students across six public universities in the mid-Atlantic region of the United States. The dataset is maintained by the Mid-Atlantic Higher Education Research Consortium (MAHERC), a research partnership formed in 2018 among the six institutions to support cross-institutional research on student success. Each member institution submits de-identified student-level records to MAHERC at the end of every academic term, and MAHERC links these records into a common student-term panel under a single IRB-approved protocol. The MAPU-SSD is well suited to our research questions for three reasons. First, all six institutions adopted EWS platforms with a common 2.0 GPA threshold, which lets us pool them in a single regression discontinuity analysis. Second, the dataset captures both the EWS flag and the downstream advisor meeting completion, which is essential for separating the effect of being flagged from the effect of receiving the intervention. Third, the six institutions include one state flagship, three regional comprehensive universities, and two minority-serving institutions, providing variation in student body composition and institutional resources that improves the generalizability of our findings. The dataset spans the 2019-20 through 2024-25 academic years, creating a six-year panel. For each enrolled undergraduate, MAPU-SSD records end-of-term cumulative GPA, term-level credits attempted and earned, registration status, declared major, demographic characteristics, financial aid status, advising appointment records, and EWS flag status. The unit of analysis is the student-term, and the dataset contains approximately 540,000 student-term observations.

Participants. Our sample consists of all first-year and sophomore undergraduates who matriculated at one of the six institutions in a fall term during our study window. We focus on lower-division students for two reasons. First, our three outcomes (next-term GPA, credit completion, and one-year retention) are measured one or two terms after the flagged term, and lower-division students still have several terms of potential enrollment ahead of them. We do not extend our sample to juniors and seniors because their situation is different. Seniors in

their final term often graduate before next-term GPA and credit completion can be measured. And for juniors and seniors close to finishing their degree, the retention measure becomes ambiguous, because a student who graduates after being flagged would appear as "not retained" in our data even though graduation is a successful outcome. Second, lower-division students are a substantively important group for studying EWS effects, since the early years of college are a period of elevated academic risk: nationally, about one in five first-year students at four-year institutions end their first year with a cumulative GPA below 2.0 (California Competes, 2025). Our sample choice is also consistent with the lower-division focus of prior regression discontinuity studies of academic probation (Fletcher & Tokmouline, 2017; Lindo et al., 2010).

Our sample includes about 45,000 lower-division students per academic year, for a total of roughly 270,000 student-term observations across the six-year panel. The demographic composition of the sample is as follows: about 56% White, 14% Black, 18% Hispanic/Latino, 7% Asian, and 5% other or multiracial; about 53% identify as female and 47% as male. About 32% are first-generation college students, and about 40% receive a Pell Grant in any given year.

EWS Implementation. All six institutions implemented their EWS platforms during the 2018-19 academic year, and our analysis window begins in 2019-20 to ensure that every student-term observation is drawn from a fully implemented system. At the end of every fall and spring semester, the system flags any student whose cumulative GPA falls below 2.0. Once flagged, a standard sequence of events follows: (a) the student is placed on academic probation; (b) a hold is placed on the student's registration for the next term; (c) the student receives an automated notification requiring a meeting with their academic advisor within four weeks; and (d) the advisor receives a parallel alert. The 2.0 cutoff is the prevailing academic-standing threshold at U.S. four-year institutions and aligns with the federal Satisfactory Academic Progress standard under Title IV of the Higher Education Act. Because all six institutions share the same cutoff, we can pool them in a single regression discontinuity analysis while controlling for institution-level differences through fixed effects.

Key Treatment Variable. Our treatment variable is the EWS probation flag, a binary indicator equal to 1 if a student's end-of-term cumulative GPA falls below 2.0 and 0 otherwise. Crossing this threshold triggers the full EWS intervention sequence described above. The forcing variable is the cumulative GPA itself, a continuous measure ranging from 0.0 to 4.0. Because cumulative GPA reflects grades across multiple courses graded by independent instructors, students cannot precisely control which side of the 2.0 cutoff they fall on. This makes the cutoff a source of quasi-random variation in treatment assignment among students near 2.0.

Advisor Meeting Completion. For Question 2, we also track whether each flagged student completed the mandatory advisor meeting within four weeks. This variable is drawn from each institution's advising appointment scheduling system and is coded as 1 if the student completed the meeting and 0 if they did not. Not all flagged students complete this meeting. Some ignore the notification, others schedule but do not attend, and still others have the hold lifted through an administrative exception. In our 2SLS analysis for Question 2, the EWS probation flag will serve as the instrument and advisor meeting completion will serve as the question predictor. The fact that compliance with the meeting requirement is incomplete is what makes it possible to separate the effect of being flagged from the effect of actually meeting with an advisor.

Outcome 1: Next-Term GPA. Our first outcome is the GPA a student earns in the term immediately following the term in which the EWS flag was triggered. This is a continuous variable ranging from 0.0 to 4.0.

Outcome 2: Credit Completion. Our second outcome is the number of credits a student passes in the term following the flagged term, where passing means earning a grade of D or higher. This variable typically ranges from 0 to 18, depending on the institution's standard course load and the student's program. We include credit completion as a separate outcome because a student's GPA can improve simply by taking fewer or easier courses, while credit completion captures whether students are also making progress toward their degree.

Outcome 3: One-Year Retention. The third outcome variable is a binary indicator equal to 1 if the student is still enrolled two terms after the flagged term, and 0 otherwise. We include retention because prior research has found that academic probation can improve short-term GPA while simultaneously discouraging some students from re-enrolling (Lindo et al., 2010). By measuring retention separately, we can test whether the EWS intervention helps students stay in school, not just earn better grades while they are enrolled.

Control Variables. We include the following pre-treatment covariates to improve the precision of our estimates: SAT or ACT score at college entry (standardized within institution), gender, race/ethnicity, first-generation status, Pell Grant eligibility, age at entry, declared major at entry, and class standing at the time of the

flagged term. We also include institution fixed effects to absorb time-invariant differences across the six universities, such as grading norms, student body composition, and advising resources. We include term fixed effects to absorb shocks shared by all institutions in a given semester, such as changes in federal financial aid policy or economic conditions.

III. Analysis

Question 1: Effect of EWS-Triggered Probation on Academic Performance and Retention

We will use a regression discontinuity (RD) design to estimate the effect of being placed on probation through the EWS. The logic is straightforward. Whether a student is placed on probation is determined entirely by whether their cumulative GPA falls below 2.0. A student with a GPA of 1.99 is flagged; a student with a GPA of 2.01 is not. These two students are almost identical in academic ability since the difference between them is a few hundredths of a GPA point, but one receives the full probation intervention and the other does not. Comparing outcomes for students just below and just above the 2.0 cutoff therefore lets us estimate the causal effect of probation at the cutoff.

Our main specification is:

$$Y_i = \beta_0 + \delta \cdot \text{Probation}_i + \beta_1 \cdot (\text{GPA}_i - 2.0) + \gamma X_i + \lambda_s + \tau_t + \varepsilon_i$$

Each observation i is a student-term in which the student's cumulative GPA was evaluated against the 2.0 cutoff. Y_i is one of our three outcomes, measured after the flagged term: next-term GPA and credit completion at the following term, and retention two terms later. Probation_i equals 1 if the student's cumulative GPA at the end of the flagged term falls below 2.0 and 0 otherwise. $(\text{GPA}_i - 2.0)$ is that same cumulative GPA recentered at 2.0. X_i is our vector of pre-treatment covariates, λ_s and τ_t are institution and term fixed effects, and ε_i is the error term. The coefficient we care about is δ. It captures the jump in the outcome at the cutoff and tells us the effect of being placed on probation by the EWS.

A regression discontinuity approach works best here for three reasons. First, treatment assignment depends only on the forcing variable and the cutoff: every student below 2.0 gets flagged, and every student above does not. Second, the 2.0 cutoff is set by institutional and federal policy, not by students themselves. Third, students cannot precisely control which side of 2.0 they end up on, because cumulative GPA reflects grades across multiple courses graded by independent instructors.

Bandwidth. For our main analysis, we include students within ±0.3 GPA points of the cutoff, that is, students with cumulative GPAs between 1.7 and 2.3. We choose this bandwidth to keep the comparison group close enough to the cutoff that students on either side are similar in academic ability, while still retaining enough observations to estimate the effect with reasonable precision.

Robustness checks. We will run three checks. First, we will re-estimate the model using polynomial and interaction specifications of the forcing variable, since RD estimates can be sensitive to functional form assumptions. Second, we will report results using ±0.2 and ±0.4 bandwidths in addition to ±0.3, since RD estimates can also be sensitive to bandwidth choice. Third, we will estimate the RD specification separately for first-year and sophomore students to test whether the effect of probation differs by class year.

Question 2: Effect of Completing the Advisor Meeting

Q1 estimates the effect of being flagged by the EWS, but the flag is only an offer of intervention. The actual receipt of services is the advisor meeting itself, and not every flagged student attends. Whether a flagged student actually attends is therefore not random: more motivated students are more likely to attend, and these same students are also more likely to improve their grades on their own. Because motivation is unobserved in our administrative data and cannot be directly controlled for, it ends up in the residuals, and meeting completion may be endogenous. We address this with two-stage least squares (2SLS), using the EWS probation flag as our instrument for advisor meeting completion. We restrict the Q2 analysis to the same ±0.3 GPA bandwidth used in Q1, so that the students in our 2SLS sample are near the cutoff and similar in academic ability.

The first stage predicts whether a flagged student actually attends the meeting:

$$\text{Meeting}_i = \gamma_0 + \gamma_1 \cdot \text{Probation}_i + \gamma_2 \cdot Z_i + \delta_i$$

The second stage uses these predicted values to estimate the effect of the meeting on outcomes:

$$Y_i = \beta_0 + \beta_1 \cdot \hat{\text{M}}\text{eeting}_i + \beta_2 \cdot Z_i + \varepsilon_i$$

Probation_i is our instrument, equal to 1 if the student was flagged by the EWS and 0 otherwise. Meeting_i equals 1 if the student completed the meeting and 0 otherwise. Z_i is our vector of pre-treatment covariates (the same as in Q1) plus institution and term fixed effects. $\hat{\text{M}}\text{eeting}_i$ is the predicted probability of meeting attendance from the first stage. Y_i is one of our three outcomes from Q1. The coefficient β_1 tells us the effect of completing the advisor meeting on outcomes for students whose attendance was changed by the flag.

A 2SLS approach will work well for our research question for three reasons. First, the EWS flag strongly predicts whether a student attends the advisor meeting, since the meeting is mandatory and a registration hold enforces attendance. We will confirm the strength of this relationship with the first-stage F-statistic. Second, near the 2.0 cutoff, whether a student is flagged is essentially quasi-random, so the flag is unrelated to motivation or other unobservables that might affect outcomes. Third, the flag plausibly affects outcomes only through the meeting itself, since there is no obvious reason why a label of "on probation" would change a student's grades except through the advising and support that the meeting provides. We will examine whether the reduced-form effect of the flag on outcomes is roughly proportional to the first-stage effect on meeting completion, which is what we would expect if the meeting is the primary channel.

Robustness checks. We will run two checks. First, we will report the reduced-form effect of the flag directly on outcomes, alongside the first-stage effect of the flag on meeting completion, to confirm that the 2SLS estimate is consistent with both. Second, we will compare the 2SLS estimate from Q2 to the RD estimate from Q1: if the meeting is the active ingredient in the EWS intervention, the 2SLS estimate should be larger in magnitude than the RD estimate.

Question 3: Heterogeneity by Race/Ethnicity and Gender

Beyond the average effect, we want to know whether EWS-triggered probation has different effects for different student groups. We test heterogeneity along two dimensions: race/ethnicity (White, Black, Hispanic/Latino, and Asian) and gender (female and male).

We do this in two ways. First, we re-estimate the Q1 RD specification separately for each subgroup, then compare the estimated δ across groups. This gives us a clean, group-specific estimate of the effect of probation. Second, we estimate a pooled specification with interaction terms between the probation indicator and each demographic variable:

$$Y_i = \beta_0 + \delta \cdot \text{Probation}_i + \Sigma_k \, \theta_k \cdot \text{Probation}_i \cdot D_{k,i} + \beta_1 \cdot (\text{GPA}_i - 2.0) + \gamma X_i + \lambda_s + \tau_t + \varepsilon_i$$

where $D_{k,i}$ are indicator variables for the demographic groups listed above. The coefficients θ_k tell us whether probation has a different effect for each group relative to the reference group. The split-sample approach gives us cleaner group-specific estimates, while the pooled specification gives us a direct statistical test of whether group differences are significant.

We acknowledge that splitting the sample by subgroup reduces the number of students near the cutoff, which reduces statistical power. We may not be able to detect moderate differences even if they exist, and we will interpret subgroup results with this limitation in mind.

The results of this analysis matter for the institutions in our sample. If the effect of probation is smaller for historically underserved students, that would raise questions about whether the current EWS design needs to be adapted. If the effect is larger for these students, that would suggest the EWS is helping to narrow attainment gaps. Either way, the answer informs how universities should design and target probation interventions going forward.

Summary. Taken together, the three analyses address the questions that a university administrator would need answered before investing in or expanding an Early Warning System. Q1 asks whether the system works at all. Q2 asks whether the advisor meeting is the part that works, or whether the flag alone is enough. Q3 asks whether it works equally well for students of different racial/ethnic backgrounds and genders. We believe this combination of a regression discontinuity design and an instrumental variables approach, applied to a multi-institution dataset, can provide the kind of evidence that is missing from current research on early warning systems in higher education.

Thank you very much for taking the time to read this proposal. We hope you will consider funding this research.